INCH-POUND
MIL-DTL-24643/69B
27 November 2023

SUPERSEDING
MIL-DTL-24643/69A(SH)
w/INT. AMENDMENT 1
11 July 2011
MIL-DTL-24643/69A
28 January 2010

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 300 VOLTS, TYPE LSC264

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Watertight)

First	-	Solid copper conductor, uncoated or coated, in accordance with ASTM B3 or ASTM B33, 0.068±0.001-inch diameter.
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation, white or natural, 0.176±0.004-inch diameter.
Fourth	-	Braided shield of 36 AWG copper, coated or uncoated. Minimum 92.5 percent coverage.
Fifth	-	Shield insulation of two polyester tapes, outer tape sealed. An alternate shield insulation may consist of one polyester tape plus an extruded jacket of a suitable transparent material; 0.003-inch minimum average thickness of the transparent material. Standard identification code shall be applied by method 2 on the inner tape.
Sixth	-	Four coaxial components shall be cabled together with a 6±1-inch, left-hand lay. Cabling sequence shall be consecutive. Fillers may be used as necessary to form a firm, well-rounded assembly.
Seventh	-	An optional binder.
Eighth	-	Braided shield of 34 AWG tin-coated copper. Minimum 90 percent coverage.
Ninth	-	An optional binder.
Tenth	-	Cross-linked polyolefin jacket. Absolute minimum average wall thickness 0.070 inch. See table I for cable overall diameter.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

Distribution Statement A. Approved for public release: distribution is unlimited.

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TABLE I. Details.

Military part no. M24643/69	Type and size	Overall diameter (inch)	
		min.	max.
-01UO	LSC264-4	0.735	0.765

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	2.5
Voltage withstand (volts, root mean square, min.)	
Conductor to shield	5,000
Shield to shield (component shields)	500
Shield to shield (component shield to overall [O/A] shield)	200
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to shield	500
Shield to shield (component shields)	100
Shield to shield (component shield to O/A shield)	100
Jacket flaws	No failure
Conductor and shield continuity	No failure

Group A:

Visual and dimensional	No failure
Mutual capacitance (nominal at 1 kilohertz [kHz]) (picofarad [pF]/100 feet)	42
Characteristic impedance (ohms) (Time Domain Reflectometer method in accordance with MIL-DTL-17)	40±2
Radio frequency transmission loss (at 400 megahertz, max.) (measure in accordance with MIL-DTL-17)	12 dB/100 feet
Watertightness (see MIL-DTL-24643 for limits for water leakage)	No failure

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Group B:	
Thermoset proof test (percent, max.)	
Jacket (when tested at 392 °F [200 °C])	50
Insulation	50
Cold bending (cable) (9-inch max. mandrel diameter)	No failure
Drip (203±1.8 °F [95±1 °C])	No failure
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Group C:	
Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Shrinkage	No failure
Heat distortion (percent of unaged, max.)	30
Permanence of printing (jacket) (cycles, min.)	125
Cable fill (sealant) removability	No failure
Shield (conformance to material, construction, and coverage)	No failure
Surface transfer impedance	
Milliohms per meter (max.)	Not applicable
Electromagnetic pulse (EMP) response (dB, min.)	Not applicable

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Group D:

Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Temperature humidity inspection (test in accordance with MIL-DTL-24643)	No failure
Dimensional stability (cable) (194±3.6 °F [90±2 °C]) (test in accordance with MIL-DTL-17)	
Inner conductor from coax core (inch, max.)	0.062
Inner conductor from coax jacket (inch, max.)	0.125
Abrasion resistance (overall jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Shield insulation	2
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (overall jacket) – type A (hardness, min.)	80

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Weathering (overall jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSC264-4	500

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-085)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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